

The Complete Love-Mode Threshold Ladder for Kirchhoff's Uniform Elliptic Vortex

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Abstract

For a uniform-vorticity ellipse in the unbounded plane, we separate the classical input from a complete all-mode consequence. The patch is an exact Euler relative equilibrium with shape rate $\Omega = \omega_0 ab / (a + b)^2$. Love's dispersion relation factors into two dimensionless scalar terms: one is positive and the other has exactly one zero for every mode $m \geq 3$. We prove strict ordering of all thresholds, the sharp first wall $a/b = 3$, and an exact closed asymptotic description of the ladder. Exact rational evidence audits 35,904 modal cells. Every stability statement is spectral and linear; no nonlinear stability, Jordan, or filamentation theorem is asserted.

1 The exact Euler relative equilibrium

Consider the planar Euler vorticity equation

$$\partial_t \omega + u \cdot \nabla \omega = 0, \quad \nabla \cdot u = 0, \quad \text{curl } u = \omega, \quad (1)$$

with velocity decaying modulo its circulation field at infinity. Let

$$E_0 = \left\{ (x, y) : \frac{x^2}{a^2} + \frac{y^2}{b^2} < 1 \right\}, \quad a \geq b > 0, \quad \omega(x, 0) = \omega_0 \mathbf{1}_{E_0}. \quad (2)$$

Positive ω_0 is counterclockwise.

Theorem 1 (Kirchhoff shape clock). *For $a > b$, the Kirchhoff patch evolution from (2) is the relative equilibrium*

$$E_t = R_{\Omega t} E_0, \quad \Omega = \frac{\omega_0 ab}{(a + b)^2}. \quad (3)$$

The interior velocity at the displayed orientation is

$$u(x, y) = \frac{\omega_0}{a + b} (-ay, bx). \quad (4)$$

The area, circulation, and quadratic vorticity moment are

$$|E_t| = \pi ab, \quad \Gamma = \pi \omega_0 ab, \quad I := \int_{\mathbb{R}^2} |x|^2 \omega(x, t) dx = \frac{\pi \omega_0 ab (a^2 + b^2)}{4}. \quad (5)$$

If $\Omega \neq 0$, the unmarked patch has minimal period $\pi/|\Omega|$; an oriented-axis lift has period $2\pi/|\Omega|$.

Proof. With $u = (-\psi_y, \psi_x)$, the interior streamfunction is

$$\psi_{\text{in}} = \frac{\omega_0}{2(a + b)} (bx^2 + ay^2),$$

which gives (4), zero divergence, and $\text{curl } \omega_0$. In elliptic coordinates $x = c \cosh \xi \cos \eta$, $y = c \sinh \xi \sin \eta$, the harmonic exterior expression

$$\psi_{\text{out}} = \frac{\omega_0 ab}{2} \xi + \frac{\omega_0 ab}{4} e^{-2\xi} \cos 2\eta + C$$

matches the boundary Cauchy data for a suitable constant C ; this is the classical Kirchhoff construction [2]. At $(a \cos \theta, b \sin \theta)$, dotting the difference between (4) and $\Omega(-y, x)$ with the unnormalized normal $(\cos \theta/a, \sin \theta/b)$ gives

$$\sin \theta \cos \theta \left\{ \frac{\omega_0(a-b)}{a+b} - \frac{\Omega(a^2-b^2)}{ab} \right\}.$$

It vanishes exactly for (3); tangential differences only reparametrize the contour. The contour therefore rotates without changing shape. Direct ellipse integration gives (5). Finally, $R_\pi E_0 = E_0$ while no smaller positive rotation fixes a noncircle, proving the two period statements. $\square$

2 Love factorization and every finite-mode wall

This revision is the **all-mode spectral-wall owner**. Put

$$\gamma = \frac{a}{b} \geq 1, \quad \delta = \frac{a-b}{a+b} \in [0, 1), \quad \kappa = \frac{2ab}{(a+b)^2} = \frac{1-\delta^2}{2}. \quad (6)$$

Proposition 1 (Sourced Love dispersion). *Resolve the boundary perturbation in elliptic coordinates whose axes co-rotate with the unperturbed ellipse. Thus the Fourier label $m \geq 1$ is measured relative to the instantaneous principal axes, and λ_m is the co-rotating-frame frequency under the convention $e^{-i\lambda_m t}$. The linearized contour equation has characteristic square*

$$\lambda_m^2 = \frac{\omega_0^2}{4} \left\{ \left[\frac{2mab}{(a+b)^2} - 1 \right]^2 - \left(\frac{a-b}{a+b} \right)^{2m} \right\}. \quad (7)$$

Equation (7) is classical Love theory, not a newly derived linearization here [1, 2]. The results below are exact consequences of this sourced characteristic equation.

Define

$$F_m(\delta) = \frac{m}{2}(1-\delta^2) - 1 - \delta^m, \quad G_m(\delta) = \frac{m}{2}(1-\delta^2) - 1 + \delta^m. \quad (8)$$

Theorem 2 (Complete finite-mode threshold theorem). *The symmetry modes obey*

$$\lambda_1^2 = \Omega^2, \quad \lambda_2^2 = 0 \quad \text{for every ellipse.} \quad (9)$$

For each $m \geq 3$, F_m has a unique zero $\delta_m \in (0, 1)$, while $G_m(\delta) > 0$ on $[0, 1)$. For $\omega_0 \neq 0$, mode m is oscillatory when $\delta < \delta_m$, critical at equality, and has an exponentially growing/decaying spectral pair when $\delta > \delta_m$.

The thresholds are strictly ordered:

$$\delta_3 < \delta_4 < \dots, \quad \gamma_m := \frac{1+\delta_m}{1-\delta_m}, \quad \gamma_3 < \gamma_4 < \dots. \quad (10)$$

The first wall is sharp:

$$\delta_3 = \frac{1}{2}, \quad \gamma_3 = 3. \quad (11)$$

Thus all $m \geq 3$ squares are positive for $1 \leq \gamma < 3$; at $\gamma = 3$ only $m = 3$ reaches zero, and for $\gamma > 3$ the $m = 3$ square is negative. These are spectral statements; zero or repeated characteristic roots are not a claim about Jordan structure.

Proof. Equations (6) and (7) give, without dividing by the possibly zero vorticity,

$$4\lambda_m^2 = \omega_0^2 F_m(\delta) G_m(\delta). \quad (12)$$

Substitution of $m = 1, 2$ proves (9). For $m \geq 3$,

$$F_m(0) = m/2 - 1 > 0, \quad F_m(1) = -2, \quad F'_m(\delta) = -m\delta - m\delta^{m-1} < 0$$

on $(0, 1)$, proving existence and uniqueness. The other factor satisfies

$$G'_m(\delta) = m\delta(\delta^{m-2} - 1) < 0, \quad G_m(1) = 0, \quad (13)$$

so $G_m > 0$ on $[0, 1)$. This proves the sign classification.

At every $0 \leq \delta < 1$,

$$F_{m+1}(\delta) - F_m(\delta) = \frac{1-\delta^2}{2} + \delta^m(1-\delta) > 0. \quad (14)$$

Hence $F_{m+1}(\delta_m) > 0$; strict decrease places its zero to the right of δ_m . The map from δ to γ is increasing. Finally, direct factorization gives

$$16\lambda_3^2 = \omega_0^2(1-\delta^2)^2(1-4\delta^2), \quad (15)$$

which proves (11) and the global first-wall statement. $\square$

References

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